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LV CABLE SELECTION AND SIZING

(600 V AND BELOW)

REVISIONS

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TRANSMISSION ENGINEERING STANDARD

TABLE OF CONTENTS

1.0 SCOPE

2.0 GENERAL

3.0 VOLTAGE RATING

4.0 CABLE CLASSIFICATIONS

4.1 Low Voltage Power Cables
4.2 Control Cables
4.3 Instrumentation Cables

5.0 CABLE SELECTION

5.1 Conductor
5.2 Insulation
5.3 Shielding
5.4 Jacket

6.0 CABLE SIZING

6.1 Ampacity
6.2 Voltage Drop
6.3 Short Circuit Capability
6.4 Miscellaneous Cables

7.0 CABLE SIZING METHODOLOGY

7.1 Criteria
7.2 Rating Correction Factors
7.3 Calculations

8.0 BIBLIOGRAPHY

APPENDIX I : SAMPLE LV CABLE SIZING CALCULATIONS
APPENDIX II : READY RECKONER FOR LV CABLE SELECTION
APPENDIX III : AMPACITY AND CORRECTION FACTORS TABLES

1.0 SCOPE

This Engineering Standard provides the design guides for the application of low voltage power, control and instrumentation cables (600 V and below, phase conductor to ground) to be used in transmission substations of Saudi Electricity Company (SEC) system.

2.0 GENERAL

2.1 The service life of the cable shall be at least equal to the design life of the substation, i.e. approximately 30 years.

2.2 LV cables shall be without splices.

2.3 Cables shall be suitable for all environmental conditions that occur in the areas where they are installed.

2.4 The cable shall be thermally stable, moisture resistant (non-hygroscopic), chemical resistant, immune to fire propagation and mechanically strong.

2.5 No single cable characteristics shall be over emphasized to the serious detriment of others. A judicious balance of cable characteristics, engineering design and good installation practice is necessary to provide a reliable cable system.

2.6 It is desirable that the system be designed so that additions and changes can be made with ease, economy and minimum outages.

3.0 VOLTAGE RATING

The rated voltage of a cable is the reference voltage for which the cable is designed. The rated voltage is expressed by the combination of two values V_O and V, expressed in volts.

V_o/V indicates for power cables under three phase current systems ($V = \sqrt{3}\,V_o$);

V_o/V indicates for control cables under single phase current and DC systems (both phase conductors insulated, $V = 2V_o$);

V_o/V_o indicates for instrumentation cables under single phase current and DC systems (one phase conductor grounded).

Whereas,

V_o signifies between any phase conductor and ground, expressed in root mean square (rms) value.

V signifies between any two phase conductors of a multicore cable or of a system of single core cables, expressed in root mean square (rms) value.

4.0 CABLE CLASSIFICATIONS

Cables covered in the scope of this standard are classified on the basis of functional requirements and thus the voltage rating for the transmission substations of SEC system.

4.1 Low Voltage Power Cables

These are designed to supply operating power to utilization devices of the substation auxiliary systems. As the three phase auxiliary system voltage is below 600 V, a voltage rating of 600/1000 V for Low Voltage (LV) power cables shall be provided.

4.2 Control Cables

These are applied at relatively low current levels or used for intermittent operation to change the operating status of a utilization device of the substation auxiliary systems. Keeping in view the single phase AC and DC system voltage values, selection uniformity and the manufacturing constraints, a voltage rating of 600/1000 V for control cables shall be provided.

4.3 Instrumentation Cables

These are multiconductor cables which transmit low energy electrical signals like variable current or voltage signals (analog) or coded information (digital). These cables are applied for SCADA digital and analog input signals, Sequence of Events Recorder, Annunciators and status/alarm signaling circuits. Due to the low voltage level of the systems in which the above transmission takes place, a voltage rating of 300/300 V or 600/600 V for instrumentation cables shall be provided.

5.0 CABLE SELECTION

When selecting cables, the rated voltage, ampacity requirements, operating and environmental conditions, safety requirements and economic considerations have to be taken into account.

5.1 Conductor

Conductor of the cables are identified with the material (copper or aluminum), type (solid or stranded) and finish (plain or tinned) as specified in IEC 60228.

Generally, copper is preferred to aluminum because of the better termination reliability and higher ampacity of copper over aluminum. Stranding of the conductor depends on the flexibility requirement. Wherever protection from fumes, gasses and other environmental conditions is required, tinned conductors shall be provided.

5.1.1 The conductor material for low voltage power cables shall be annealed copper or aluminum. The conductors up to 630 mm² size shall be stranded.

While the conductors of 1.5 and 2.5 mm^2 sizes shall be either solid or stranded.

5.1.2 Control cable conductor material shall be annealed copper. Conductors of multicore control cable shall be of size 1.5 mm² (excluding current transformer circuit) and stranded as well. The cable size for current transformer secondary circuit shall be minimum 2.5 mm² as specified in clause 6.4.3.

5.1.3 Instrumentation cable conductor material shall be annealed copper, either solid or stranded. The cable size shall be 0.5 mm² up to 2.5 mm^2 depends on application criteria required for the design of instrumentation cable.

5.2 Insulation

The insulating material for conductor of all types of cable covered in this standard shall be Polyvinyl Chloride (PVC) or cross-linked Polyethylene (XLPE). Normally, PVC insulation having a maximum operating temperature of 85°C with Limiting Oxygen Index (LOI) of 28 - 32 % per ASTM D2863 shall be provided. However, if better dielectric strength or thermal stability is required, XLPE insulation having a maximum operating temperature of 90° with LOI of 20 – 23 % shall be provided as conductor insulation. LOI of an insulated cable is a measure of fire propagation resistance of the insulating material. Air normally contains 21 percent oxygen, hence, a material with oxygen index below 21 will burn easily in air.

5.3 Shielding

Shielding of cable is for the purpose of reducing or eliminating electrostatic interference between conductors or groups of conductors within the cable or outside interference on cable conductors. All the control cables for use with static relays, current transformer secondary circuit, voltage transformer secondary circuit and instrumentation cables shall be provided with metallic shield of flat bare copper tape, helically applied.

5.4 Jacket

Jacket (Sheath) is used over individual insulated wires or over multiconductor cables to provide environmental criteria, mechanical protection and fire resistance. Extruded PVC, ST_2 type being relatively lower in cost, non-hygroscopic and inherently flame retardant with LOI of 28 - 30 % shall be provided as sheathing material.

6.0 CABLE SIZING

Ampacity (current carrying capacity) is the important element in cable sizing. Ampacity will set the minimum size of conductor that can be used for a given type of application.

Voltage drop is the second factor, considered to check whether the selected cables are capable of supplying proper operating voltage range. The allowable voltage drop shall be limited to 5 % of the system nominal voltage throughout, from the supply sources to the load ends. Voltage drop restriction will require an increase in conductor size.

Short circuit current will require an increase in conductor size selected if it is not sufficient to dissipate the heat generated under fault conditions. In substation LV cable systems, conductor short circuit capability shall not be considered. However, cable from station transformer to main distribution board shall be checked for short circuit capability.

6.1 Ampacity

Conductor sizing based on ampacity is made by calculating the current required to serve the connected load. The cable shall be capable of carrying normal and emergency load currents. The conductor size is then selected from cable manufacturer's standard ampacity tables. Minimum size of AC and DC power cables shall be 1.5 mm².

Appropriate ampacity correction (derating) factors and adjustment formula for change in parameters shall be applied in using the ampacity tables for various cable constructions and methods of installation given in the applicable industry standard or manufacturer's data.

Where cables are routed through several types of installation conditions, conductor size shall be selected based on the most severe site and environmental conditions.

6.2 Voltage Drop

When a circuit supplies power to a load, it experiences a drop in voltage and a dissipation of energy in the form of heat. With cables in the DC systems and cables with cross sectional area upto 16 mm² in AC single and three phase systems, only the DC resistance at maximum operating temperature has to be taken into account for calculating the voltage drop. With cables of more than 16 mm² installed in AC single phase or three phase systems, however, the resistance and the reactance have to be taken into account. The voltage drop, throughout the length of the circuit, can be estimated as follows :

DC : $V_{drop} = 2 \times I \times R \times L$ Volts

AC Single Phase : $V_{drop} = 2 \times I \times L \times (R \cos\phi + X_L \sin\phi)$ Volts

AC Three Phase : $V_{drop} = \sqrt{3} \times I \times L \times (R \cos\phi + X_L \sin\phi)$ Volts

Where,

I	=	Load current in Amperes
R	=	Resistance of cable in Ohms per meter
X_L	=	Inductive reactance in Ohms per meter
L	=	Length of cable route in meters

ϕ = Power factor angle

IEC 60228 provides the maximum DC resistance at 20°C of stranded conductors for single core and multicore cables which is required to be converted to the value at maximum operating temperature of the cable. The same can be calculated with the help of the following formula :

DC Resistance (R) $= R_{20} \times [1 + \alpha_{20} \times (\theta - 20)]$

Where,

R_{20} = DC resistance of conductor at 20°C in Ohms per meter (see Table II of IEC 60228)

α_{20} = Constant mass temperature coefficient at 20°C per °C
(α_{20} = 0.00393 per °C at 20°C for copper conductor)
(α_{20} = 0.00403 per °C at 20°C for aluminum conductor)

θ = Maximum operating temperature of the cable in °C

The conversion of resistance can also be arrived at with the help of the ready reckoner for the factor [1 + α_{20} (θ - 20)] as given in Table 29-7 of Appendix III. The same value of DC resistance is also applicable for AC resistance because in case of LV power cable, skin effect and proximity effect are negligible.

The other factors required for calculation of voltage drop as per above formula are reactance value of the cable and values of cos ϕ and sin ϕ. Reactance value is provided by the cable manufacturer. The values of cos ϕ and sin ϕ can be obtained from connected load.

However, manufacturers of the cable, in general, provide the value of DC resistance at 20°C, AC resistance at maximum operating temperature and reactance for different sizes of cable directly in their ampacity tables.

Apart from above formula, voltage drop can also be calculated directly with the help of readily available unit voltage drop values in millivolts per ampere per meter at 70°C as per Table 4D IB and 4D IIB of BS 7671, 1992. These values of unit voltage drop can be converted to conductor maximum operating temperature of 85°C with the help of Table 29-7 and Table 29-8 of Appendix III. A ready reckoner is provided for PVC insulated cables at 85°C in Table 29-9, of Appendix III.

The formula for voltage drop calculation shall be :

$$V_{drop} = \frac{I \times U \times L}{1000} \quad \text{Volts}$$

Where,

I	=	Load current in Amperes
U	=	Unit voltage drop in millivolts per Ampere per meter
L	=	Length of cable route in meters

After finding out the voltage drop, calculation for voltage regulation shall be made with the following formula:

$$V_e = \frac{V_{drop}}{V_S} \times 100 \qquad (\%)$$

Where,

V_e	=	Voltage regulation in percent
V_{drop}	=	Total voltage drop in volts
V_s	=	Supply voltage at source in volts

6.3 Short Circuit Capability

Short circuit capability of the cable size selected for station transformer to main distribution board shall be checked with the formula derived on the basis of IEC 60724.

$$I_{sc} = \frac{0.1038}{\sqrt{t}} \times A \qquad \text{(Copper Conductor PVC insulated)}$$

$$I_{sc} = \frac{0.143}{\sqrt{t}} \times A \qquad \text{(Copper Conductor XLPE insulated)}$$

Where,

I_{sc}	=	Short circuit current in kiloamperes
t	=	Duration of short circuit current in seconds
A	=	Cross sectional area of conductor in mm^2

The short circuit capability of a particular cable size can also be checked directly from the short circuit capability graph for the cables, provided by the manufacturers (see Figure 29-7 of Appendix III).

6.4 Miscellaneous Cables

6.4.1 Battery power cables from batteries to DC main distribution board (DCDB) for substations shall be sized to handle the heavy current (momentary discharge) associated with the battery bank or batteries and to minimize voltage drops. Battery power cables shall be oversized by a safety factor of

25 % to assure reliability. The cable shall also be checked for short circuit capability.

6.4.2 Control cables are applied at relatively low current levels. The cross sectional area of the conductors are generally small and the duty cycle is such that conductor heating is insignificant. Voltage drop in control cables shall be limited to 5 % of the source voltage, to insure satisfactory operation of coil devices.

6.4.3 Current transformer (CT) secondary circuit cables shall be sized on the basis of knee point voltage limit (V_k). Voltage developed (V_d) across the secondary circuit at the time of fault depends upon the connected burden. The CT with certain knee point voltage and secondary winding resistance and also relays and instruments having specific burden are arranged in advance to cable sizing. It is only the cable burden which can be selected according to the CT requirements. The cable size is so selected that V_d remains lower than the knee point voltage and the ratio error within the acceptable limit.

It is important that CT secondary cable sizing is done as a part of CT sizing calculations.

Voltage developed across secondary circuit of current transformer to drive the ratio current during a fault shall be computed by the following typical formula or the specific formula recommended by the relay manufacturer for a particular protection :

$$V_d = I_F (R_{CT} + R_L + R_R) \quad \text{Volts}$$

Where,

I_F = Fault current converted to CT secondary
R_{CT} = CT secondary winding resistance of the core tap used
R_L = Two times CT secondary one way lead resistance
R_R = Relay resistance

In case of delta connected CTs for differential protection, R_L shall be considered as three times the one way lead resistance for above computation.

Standard control cable for CT circuits shall be 600V, 4-core, stranded copper. CT secondary circuit wire of size 2.5 mm² minimum shall be provided. Standard insulation shall be XLPE and standard sheath shall be PVC.

6.4.4 Capacitor voltage transformer (CVT) circuit cables shall be selected on the basis of durability and quality of possible connections.

Standard control cable for CVT circuits shall be 600 V, 4-core stranded copper. In no case, wire smaller than 2.5 mm² for CVT shall be provided. Standard insulation shall be XLPE and standard sheath shall be PVC.

6.4.5 Instrumentation cables are supposed to carry milliamperes, analog or digital signals at low voltage levels, their sizing is not of much significance. However, in view of mechanical reliability, stranded copper conductor of size 0.8 mm² shall be provided.

7.0 CABLE SIZING METHODOLOGY

7.1 Criteria

For calculation of LV power cable conductor size, the relevant IEC, ICEA and NEC guidelines shall be followed. However, the following technical particulars shall be the criteria for all the calculations of conductor size.

7.1.1 Outdoor cables shall be laid in underground concrete ducts and indoor cables in well ventilated raceways like ladders and racks. Cables from duct bank to outdoor equipment terminal box/local control panels shall be laid in Rigid Steel Galvanised (RSG) conduit. A clearance of minimum 50 mm shall be maintained from ground, ceiling and wall and a clearance of 300 mm between the ladders/racks shall also be maintained.

7.1.2 Maximum air ambient temperature

a.	Indoor	40°C
b.	Outdoor	50°C

7.1.3	Maximum ground temperature	40°C
7.1.4	Depth of burial in ducts shall be	750 mm
7.1.5	Soil thermal resistivity (concrete duct bank : 1.2°C. m/W)	2.5°C. m/W

7.1.6 Conductor temperature for maximum continuous rating shall be considered 85°C for PVC and 90°C for XLPE insulated cables.

7.1.7 Grouping correction factor as below :

a. Cables from station transformer to main distribution board :

i. 0.87 for single core cable with trefoil touching configuration
ii. 0.82 for single core with flat touching configuration
iii. 0.82 for multicore cable with flat touching configuration

b. Cables from main distribution board to various loads :

i. 0.45 for cable up to 16 mm² (to be laid double layer touching)
ii. 0.70 for cable 25 - 70 mm² (to be laid single layer touching)
iii 0.70 for cable above 70 mm² (single layer with 25 mm spacing)

7.1.8 Power factor of motor, continuous operation condition, cos ϕ = 0.85.

7.1.9 Voltage drops (regulation in %) shall be limited as :

a.	Station transformer to main distribution board (MDB)	0.8%
b.	MDB to various loads (Motor starting voltage drop limited to 12.5%.)	4.2%
c.	Battery to DC distribution board (DCDB)	0.8%
d.	DCDB to various loads	4.2%
e.	DCDB to control/relay panel	0.4%
f.	Control/relay panel to DC loads	3.8%

Total voltage drop from station transformer and battery to the load end shall not exceed 5 %. However, in case of circuit breaker trip coils which are rated for 70 - 140V, the voltage drop limit shall be considered upto 10 %.

All DC voltage drop shall be calculated below end of discharge voltage which is 101.5 V in case of 125 V battery.

Since the different cable manufacturers evaluate the conductor ampacity on varying criteria, as such, they shall be asked to provide the various parameters considered for conductor ampacity and necessary correction factors as well, based on guidelines of IEC 60228 and IEC 60287 (applicable parts).

7.2 Rating Correction Factors

The rating correction factors (derating factors) shall depend on the nature of cable laying and manufacturer's cable ampacity table criteria. The formula for total rating correction factors calculation shall be as follows:

$$K_{tot} = K_1 \times K_2 \times K_3 \times .... K_n \ , \ n = 1, 2, 3, ... \text{integer number}$$

Where,

K_{tot} = Total rating correction factor

K_1 ; K_2 ; K_3 ; … K_n = Partial rating correction factor, as many as the differences of laying and operating conditions from standard conditions, as listed below

Various correction factors applied for cable sizing are comprehensively listed below:

7.2.1 Cable laid in Air (Ventilated Ladders/Racks)

a. Air Temperature Correction Factor

i. PVC insulated cable
ii. XLPE insulated cable

b. Grouping Correction Factor

i. Single core cable (trefoil touching/flat touching)
ii. Multicore cable (flat touching)

7.2.2 Cable Laid in Ground (Concrete Duct)

a. Ground Temperature Correction Factor

i. PVC insulated cable
ii. XLPE insulated cable

b. Depth of Burial Correction Factor

c. Thermal Resistivity of Soil Correction Factor

d. Duct Laying Correction Factor

e. Grouping Correction Factor

i. Single core cable (trefoil touching)
ii. Multicore cable (flat touching)

Grouping correction factor for trefoil spaced and flat spaced is not considered, as the LV power cable shall not be laid in these formations.

Some of the manufacturers provide conductor current rating for trefoil touching in air, trefoil touching in ground and trefoil touching in duct readily in their conductor ampacity tables. In such cases, rating correction factors for these conditions are not required and shall be considered as one (1).

7.3 Calculations

Sizing of LV power cable for a particular load requires current rating calculation in amperes of the load. The current rating shall be subjected to various correction factors, as necessary. This corrected value of current will be applied for finding out a suitable size of conductor from the ampacity table of the manufacturer. The size, so selected, shall be checked for voltage drop. If the voltage drop exceeds the limit as per clause 7.1.9, then the size of the conductor shall be increased to have the voltage drop within the limits. Cables for motor circuits shall be checked for voltage drop against their starting current. Then the size of conductor shall be checked for short circuit capability, wherever required.

In case of current transformer cable, voltage developed across the secondary circuit during a fault shall be computed as per clause 6.4.3 for each protection individually as part of CT sizing calculation on the basis of the specific formula provided by the relay manufacturer. The voltage thus computed should be lower than the knee point voltage of the current transformer.

On the basis of above guidelines, the design engineer shall calculate the sizes of LV power cable. Control cables shall be checked for voltage drop only.

However, for reference, few sample calculations have been provided in Appendix I. Further, Appendix II provides tables and plotted graphs (self explanatory) of different loads/cable lengths and respective sizes of the various LV power and control cables, for usage in Saudi Electricity Company (SEC) substations.

Sample manufacturer cables ampacity tables, various correction factor tables, resistance/reactance table, unit voltage drop table and short circuit capability graph required for cable sizing is provided in Appendix III.

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APPENDIX I

SAMPLE LV CABLE SIZING CALCULATIONS

In the following illustrative example, few sample calculations are made involving LV power cables, control cables and current transformer secondary circuit cables.

All calculations are made on the basis of sample manufacturer ampacity tables, various correction factor tables, resistance/reactance table, unit voltage drop table and short circuit capability graph which are enclosed as Appendix III. Various ambient conditions, on which different data are based, have been specified below each and every table. Cable sizes so calculated shall also be applicable for other manufacturers. The calculations are made for PVC (85°C) insulated cables but the same sizes shall also be applicable for XLPE (90°C) insulated cables.

Detailed voltage drop calculation is made as per clause 6.2 and values of various factors of voltage drop formula provided in Appendix III.

Various correction factors and criteria are applied as per clause 7.0. Since the ampacity of the cable laid in concrete duct, under the ground, is directly available in the cable ampacity table, no correction of ampacity is applied on account of laying in duct. The depth of laying and thermal resistivity of soil considered are same as that adopted in this standard, hence no derating factor shall be applied on these accounts. Further, when the cable size goes higher than 300 mm² and bending radius is high, single core cable is preferred for the sake of ease in cable laying.

1. Cable from 500 kVA, 13.8 kV/380 V station service transformer to main distribution board (MDB)

   a. Ampacity

For three phase AC,

$$I = \frac{P}{\sqrt{3} \times V \cos\phi} \text{ A} \quad \text{(P in Watts)}$$

$$= \frac{P}{\sqrt{3} \times V} \text{ A} \quad \text{(P in VA)}$$

$$= \frac{500\text{x}10^3}{\sqrt{3}\text{x}380}$$

Maximum full load current, $I \approx 760$ A

The cable required to carry 760A, is too big in size to lay, bend and terminate and hence two cables in parallel, each carrying 380A, are required. As the size of each cable is still expected to go above 300 mm², single core cable is considered.

Estimate: 2 sets of 3 single core cables in parallel.

Correction Factors:

Since the station transformer shall be kept outdoor, part of LV cable is laid in concrete duct and hence ambient conditions of concrete duct (underground) shall be applicable.

Ground temp. correction factor = 0.95
(for ground ambient temp. of 40°C) (Table 29-6 of Appendix III)

Duct laying correction factor = 1.0
(since ampacity for laying in duct is directly available)

Depth of burial correction factor = 1.0
(since manufacturer's depth of laying is same)

Thermal Resistivity of soil correction factor = 1.0
(since thermal resistivity of soil is the same)

Group correction factor = 0.87 (clause 7.1.7.a)

Fictitious Current (A), I_f:

The formula for fictitious current (I_f) calculation shall be as follows:

$$I_f = \frac{I}{K_{tot}} = \frac{I}{K_1 \times K_2 \times K_3 \ldots \times K_n}, \; n = 1, 2, 3, \ldots. \text{etc.}$$

Where,

I_f	=	Fictitious current in amperes (A)
I	=	Continous current rating at operating conditions (A)
K_{tot}	=	Total correction factors (derating factors)
K_n	=	Partial correction factor, n=1,2,3, … integer number

As per parameters given, I = 380 A, K_1 = 0.95, K_2 = 1.0, K_3 = 1.0, K_4 = 1.0, K_5 = 0.87, the fictitious current to be carried by each set of cable:

$$I_f = \frac{380}{0.95 \times 1 \times 1 \times 1 \times 0.87} = 459.7 \approx 460 \text{ A}$$

As per Table 29-3 of Appendix III, using copper conductor, PVC insulated and PVC sheathed, the nearest cable size for carrying 460 A in ground concrete duct continuously is 400 mm².

<u>Cable particulars:</u>

2 sets of 3 x 1C x 400 mm² copper conductor, PVC insulated.

b. Voltage Drop

- By formula,

$$V_{drop} = \sqrt{3} \times I \times L \times (R \cos\phi + X_L \sin\phi) \quad \text{Volts} \qquad \text{(clause 6.2)}$$

Where,

I	=	380 A	
L	=	40 m	(typical)
R	=	0.0628 Ω/km	(Table 29-8 of Appendix III)
X_L	=	0.091 Ω/km	(Table 29-8 of Appendix III)
cos ϕ	=	0.85	(clause 7.1.8)
ϕ	=	31.8°	
sin ϕ	=	0.527	

$$V_{drop} = \sqrt{3} \times 380 \times 40 \times (0.0000628 \times 0.85 + 0.000091 \times 0.527)$$

$$= 1.73 \times 15200 \times (0.0000534 + 0.0000480)$$

$$= 1.73 \times 15200 \times (0.0001014)$$

$$= 26296 \times (0.0001014)$$

$$= 2.66 \text{ V}$$

Voltage Regulation, $V_e = \dfrac{V_{drop}}{V_S} \times 100 \quad (\%)$

$$= \frac{2.66}{380} \times 100 = 0.7\,\%$$

Voltage regulation is within the limit of 0.8 % (clause 7.1.9.a) and hence the selected cable size of 400 mm² is appropriate.

- By unit voltage drop,

$$V_{drop} = \frac{I \text{ x } U \text{ x } L}{1000} \quad \text{Volts} \qquad \text{(clause 6.2)}$$

Unit voltage drop, U = 0.19 mV/A/m (Table 29-9 of Appendix III)

$$V_{drop} = \frac{380 \text{ x } 0.19 \text{ x } 40}{1000} = 2.888 \text{ V}$$

Voltage Regulation, $V_e = \frac{V_{drop}}{V_S} \times 100$ (%) (Clause 6.2)

$$V_e = \frac{2.888}{380} \times 100 = 0.76\,\%$$

This value of voltage regulation (V_e) is same as that of voltage regulation calculated by formula. Minor difference in voltage regulation is because of power factor cos ϕ, considered as 0.7 for ready reckoner calculations.

Further calculations of voltage drop is made only with the help of ready reckoner Table 29-9 of Appendix III.

c. Short Circuit

The conductor size is checked for short circuit with the help of formula,

$$I_{sc} = \frac{0.1038}{\sqrt{t}} \times A \quad \text{(Copper Conductor PVC insulated)} \qquad \text{(clause 6.3)}$$

$$I_{sc} = \frac{0.1038}{\sqrt{1}} \times 400 = 41.5 \text{ kA}$$, for one second

and $$I_{sc} = \frac{0.1038}{\sqrt{3}} \times 400 = 24.0 \text{ kA}$$, for three seconds

Conductor size of 400 mm² selected, is capable of bearing 41.5 kA for one second and 24 kA for three seconds which is more than enough for LV bus system generally adopted in Saudi Electricity Company substations.

The conductor size was checked for short circuit with the help of Figure 29-7 of Appendix III also and found same as that calculated by formula.

Further checking of short circuit capability is made with Figure 29-7 of Appendix III only.

2. <u>300 kVA, 13.8 kV/380 V station service transformer to main distribution board (MDB)</u>

a. Ampacity

Maximum full load current for 300 kVA at 380 V, I = 456 A

Two cables, carrying 228 A each, shall be laid in parallel to have the lower size of the cable for the reasons as considered in sample calculation 1.

Estimate: 2 nos. multicore cable in parallel.

Ground temperature correction factor (K_1) = 0.95 (Table 29-6 of Appendix III)

Group correction factor (K_2) = 0.82 (clause 7.1.7.a)

The fictitious current to be carried by each cable,

$$I_f = \frac{I}{K_1 \times K_2} = \frac{228}{0.95 \times 0.82} = 292.68 \approx 293 \text{ A}$$

As per Table 29-4 of Appendix III, using copper conductor, PVC insulated and PVC sheathed, the nearest and safe cable size for carrying 293 A in ground concrete duct continuously is 240 mm².As the cable capacity is too close to the current to be carried and due to the importance of this cable which connects the station service transformer and the station loads, the next higher size of 240 mm² is selected.

Cable Particulars:

2 nos. 4C x 240 mm² copper conductor, PVC insulated.

b. Voltage Drop

Length of cable route, L = 40 m (typical)
Unit voltage drop, U = 0.21 mV/A/m (Table 29-9 of Appendix III)

$$V_{drop} = \frac{228 \text{ x } 0.21 \text{x } 40}{1000} = 1.92 \text{ V}$$

Voltage Regulation, $V_e = \frac{1.92}{380} \times 100 = 0.50\,\%$

Value of voltage regulation is within the limit of 0.8 % (clause 7.1.9.a) and therefore the selected cable size of 240 mm² is appropriate.

c. Short Circuit

The conductor size was checked for short circuit capability, using Figure 29-7 of Appendix III and found to be within acceptable limit.

3. Cable from MDB to air condensing unit

a. Ampacity

Load of air condensing unit at 380 V = 63 kVA

Full load current for 63 kVA at 380 V = 96 A

Estimate: 1 no. multicore cable.

Air temperature correction factor (K_1) = 1.16 (Table 29-5 of Appendix III)
Group correction factor (K_2) = 0.70 (clause 7.1.7.b)

Therefore fictitious current, $I_f = \frac{I}{K_1 \times K_2} = \frac{96}{1.16 \times 0.70} = 118.2 \approx 118$ A

As per Table 29-4 of Appendix III, using copper conductor, PVC insulated and PVC sheathed, the nearest cable size for carrying 118 A in underground duct continuously to motor is 50 mm².

b. Voltage Drop

The above cable size is to be confirmed only after checking the voltage drop with respect to starting current of motor of air condensing unit.

Starting current,	I	=	576 A	(six times full load current)
Length of cable route,	L	=	45 m	(typical)
Unit voltage drop,	U	=	0.85 mV	(Table 29-9 of Appendix III)

$$V_{drop} = \frac{576 \text{ x } 0.85 \text{ x } 45}{1000} = 21.95 \text{ V}$$

Adding station transformer to MDB voltage drop of 1.92 V, (section 2.b)

Total $V_{drop} = 22.0 + 1.92 = 23.92$ V

Voltage Regulation, $V_e = \frac{23.92}{380} \times 100 = 6.30$ %

Since the voltage regulation is within the limit as per clause 7.1.9.a & b, the selected cable size of 50 mm² is appropriate.

To find out the maximum length to which this size of cable can be laid for above load with the starting condition voltage drop limit of 47.5 V (12.5%), the values are put in unit voltage drop formula.

$$L = \frac{V_{drop} \times 1000}{I \text{ x } U}$$

$$L = \frac{47.5 \times 1000}{576 \text{ x } 0.85} \approx 97.0 \text{ m}$$

Thus for a motor load of 96 A, a copper conductor cable of size 1x4cx50mm² can be laid upto a maximum length of 97 meters.

Cable Particulars:

1 no. 4C x 50 mm² copper conductor, PVC insulated.

4. Cable from 125 V battery to DC distribution board (DCDB)

Battery to DCDB power cable sizing is based on voltage drop calculation, considering momentary load and continuous load. During the bus bar fault, simultaneous tripping of breakers connected to the bus results into maximum momentary load requirement from battery for trip coil operation. The load is calculated by multiplying trip coil current rating with number of trip coil per circuit breaker and number of circuit breakers to be connected to busbar.

Consider,	continuous load of the battery	=	50 A
	Momentary load of the battery	=	150 A
	Total load of the battery, I	=	200 A
	Increasing by 25% (clause 6.4.1)	=	250 A

As per Table 29-3 of Appendix III, using single core copper conductor, PVC insulated and PVC sheathed, the nearest safe cable size for carrying 250 A in air is 95 mm². This size can only be confirmed after voltage drop check with actual load of 200 A.

Length of cable route, L	=	14 m	(typical)
Unit voltage drop, U	=	0.51 mV/A/m	(Table 29-9 of Appendix III)

$$V_{drop} = \frac{200 \text{ x } 0.51 \text{ x } 14}{1000} = 1.428 \text{ V}$$

As the voltage drop is too high at source, the cable size is increased to, say 240 mm².

Unit voltage drop, U = 0.24 mV/A/m (Table 29-9 of Appendix III)

$$V_{drop} = \frac{200 \text{ x } 0.24 \text{ x } 14}{1000} = 0.672 \text{ V}$$

Voltage Regulation, $V_e = \frac{0.672}{125} \times 100 = 0.54\ \%$

Since the voltage regulation is within the limit of 0.8 % (clause 7.1.9.c), the selected cable size of 240 mm² is appropriate.

From Figure 29-7 of Appendix III, it is evident that the short circuit capability of this conductor size is more than adequate.

Cable particulars:

2 nos. 1C x 240 mm² copper conductor, PVC insulated.

5. Cable from DCDB to circuit breaker with DC motor

The basis of DC motor cable sizing is voltage drop at load current only and no consideration is made for starting current. The starting current of DC motor is very close to normal load current.

Consider, continuous load of motor, I = 15 A
Length of cable route, L = 200 m (typical)

As per Table 29-3 of Appendix III, for a load of 15 A, selecting twin core copper, PVC insulated and PVC sheathed, the nearest safe cable size for carrying 15 A in ground concrete duct is 2.5 mm². This size can only be confirmed after voltage drop check.

Unit voltage drop, U = 18.6 mV/A/m (Table 29-9 of Appendix III)

$$V_{drop} = \frac{15 \text{ x } 18.6 \text{ x } 200}{1000} = 55.8 \text{ V}$$

As the voltage drop of 55.8 V is too high, the cable size is increased to, say 35 mm².

Unit voltage drop, U = 1.3 mV/A/m (Table 29-9 of Appendix III)

$$V_{drop} = \frac{15 \text{ x } 13 \text{ x } 200}{1000} = 3.9 \text{ V}$$

Adding Battery to DCDB Voltage drop of 1.0 V,

Total voltage drop from battery to load = 3.9 + 1.0 = 4.9 V

Voltage Regulation, $V_e = \frac{4.9}{125} \times 100 = 3.92 \ \%$

Since the voltage regulation is within the limit of 5 % (clause 7.1.9.c & d), the selected cable size of 35 mm² is appropriate.

Cable Particulars:

1 no. 2C x 35 mm² copper conductor PVC insulated.

6. Cable from relay panel to trip coil of circuit breaker

Size of cable to trip coil of circuit breaker shall be so selected that the total voltage drop (from battery to DCDB, from DCDB to relay panel and from relay panel to trip coil) does not exceed the limit of 10 % (clause 7.1.9).

7. Cable from relay panel to current transformer

For a 230 kV line with a fault level of 40 kA for one second, the following typical formula for voltage developed (V_d) across CT secondary circuit is considered for CT secondary cable sizing.

$$V_{drop} = I_F \times \left(\frac{X}{R} + 1\right) \times \left(R_{CT} + R_L + R_R\right)$$

Assuming,

CT ratio		=	3000/5
CT knee point voltage	V_k	=	1080 V
CT secondary resistance	R_{CT}	=	1.0 Ω
Relay resistance	R_R	=	0.5 Ω
Two way lead length		=	150 x 2 = 300 m
Unit resistance of 4 mm² cable		=	5.79 Ω/km
Lead resistance	R_L	=	1.74 Ω
	$\frac{X}{R}$	=	4.4

$$V_{drop} = \frac{40 \times 10^3 \times 5}{3 \times 10^3} \times (4.4 + 1) \times (1.0 + 1.74 + 0.5)$$

$$= 66.66 \text{ x } 5.4 \text{ x } 3.24 \text{ V}$$

$$\approx 1166 \text{ V}$$

The calculation made above shows that if 4 mm² cable is laid between CT and relay, the voltage developed (V_d) in CT secondary circuit at the time of fault, to drive the ratio current, is 1166 V. This value of V_d is more than the knee point voltage, 1080 V of CT to which relay is to be connected and hence the CT will get saturated before the full fault current is reached. The only option to reduce the CT secondary voltage requirement is to reduce the lead burden by increasing the size of cable from 4 mm² to 6mm².

Unit resistance of 6 mm² cable		=	3.87 Ω/km
Lead resistance	R_L	=	1.16 Ω

$$V_{drop} = \frac{40 \times 10^3 \times 5}{3 \times 10^3} \times (4.4 + 1) \times (1.0 + 1.16 + 0.5)$$

$$= 66.66 \text{ x } 5.4 \text{ x } 2.66 \text{ V}$$

$$\approx 958 \text{ V}$$

As the voltage value of 958 V, required at the time of fault to drive the ratio current, is lower than the knee point voltage of the CT, to which relay is to be connected, the cable size of 6 mm² is appropriate for interconnection of relay to current transformer.

<u>Cable particulars:</u>

1 no. 4C x 6 mm² copper conductor, PVC insulated.

APPENDIX II

READY RECKONER FOR LV CABLE SELECTION

This appendix consists of Tables 29-1 and 29-2 and Figures 29-1 to 29-6 providing suitable copper conductor cable sizes (single core or multicore) for different loads (AC or DC) and length with desired voltage drop limit. The various calculations carried out were in accordance with clause 7.0 of this Standard.

Key to Figures 29-1 to 29-6

To find out a suitable size of the cable for a particular load and length, select the appropriate figure with desired voltage drop limit. Start from lowest size of cable and see if the load ampere is available on the bar or not. If it is not available, move to higher size of cable. After finding out suitable size of cable, see how much length (X-axis) is allowed for that load. If the length of the cable to be laid is within the limit of the length found from the figure, the size of cable is appropriate. If the length is more than what is found from the figure, then move to higher size of cable and check, as explained above, till the right size of cable is found.

Figure 29-1 provides cable sizes for motor load which satisfy starting voltage drop conditions also. Cable sizes shall be suitable for laying both in air and duct. The points encircled are suitable for laying in air only. Numerals mentioned above the points marked on the bar are load values in amperes.

Figure 29-2 provides multicore cable sizes for non-motor load.

Figure 29-3 provides single core cable sizes for non-motor load.

Figures 29-4, 29-5 and 29-6 provide two core cable sizes for DC power and control cables.

Table 29-1: DC Power Copper Conductor Cable for Various Loads with Different Voltage Drops as per Clause 7.1.9

Sr. No.	Route	Load (A)	Type	PVC/PVC XLPE/PVC	Actual Length of Cable (m)	Unit Voltage Drop (mV/A/m)	Actual Voltage Drop (V)	Total Voltage Drop (V)	Voltage Delivered (V)	Voltage Regulation (%)
1.	From 125 V Battery to DCDB	300	1C	2 x 1C x 400 2 x 1C x 300 2 x 1C x 240	≤ 14 ≤ 12 ≤ 10	0.19 0.21 0.24	0.79 0.75 0.72	0.79 0.75 0.72	124.21 124.25 124.28	0.63 0.60 0.58
		250	1C	2 x 1C x 300 2 x 1C x 240 2 x 1C x 185	≤ 14 ≤ 12 ≤ 10	0.21 0.24 0.29	0.74 0.72 0.72	0.74 0.72 0.72	124.26 124.28 124.28	0.59 0.58 0.58
		200	1C	2 x 1C x 240 2 x 1C x 185 2 x 1C x 150	≤ 14 ≤ 12 ≤ 10	0.24 0.29 0.35	0.68 0.70 0.70	0.68 0.70 0.70	124.32 124.30 124.30	0.54 0.56 0.56
		150	1C	2 x 1C x 150 2 x 1C x 120 2 x 1C x 95	≤ 14 ≤ 12 ≤ 10	0.35 0.41 0.51	0.74 0.74 0.76	0.74 0.74 0.76	24.26 124.26 124.24	0.59 0.59 0.61
		100	1C	2 x 1C x 95 2 x 1C x 70 2 x 1C x 50	≤ 14 ≤ 10 ≤ 7	0.51 0.69 0.98	0.71 0.69 0.69	0.71 0.69 0.69	124.29 124.31 124.31	0.57 0.55 0.55
		50	1C	2 x 1C x 50 2 x 1C x 35 2 x 1C x 25	≤ 14 ≤ 11 ≤ 8	0.98 1.3 1.8	0.69 0.72 0.72	0.69 0.72 0.72	124.31 124.28 124.28	0.55 0.57 0.57
	By keeping the voltage drop on lower size i.e. within 0.80 V instead of 1.0 V, the voltage drop at DCDB, due to the drops at cable terminations, has not been allowed to exceed 0.8% as per clause 7.1.9-c.									
2.	From 125 V DCDB to Control/Relay Panel	3	2C	1 x 2C x 10 1 x 2C x 6 1 x 2C x 4 1 x 2C x 2.5	≤ 34 ≤ 20 ≤ 14 ≤ 8	4.60 7.70 11.60 18.60	0.47 0.46 0.48 0.45	0.47+1.0=1.47 0.46+1.0=1.46 0.48+1.0=1.48 0.45+1.0=1.45	123.53 123.54 123.52 123.55	1.17 1.16 1.18 1.16
	(Sr.no.1+2 drop rounded off to 1.5 V, the value meets the criof 0.8% + 0.4% drop as per clause 7.1.9-c & e)									
3.	From Control/Relay Panel to Loads	3	2C	1 x 2C x 10 1 x 2C x 6 1 x 2C x 4 1 x 2C x 2.5	≤ 318 ≤ 190 ≤ 126 ≤ 80	4.60 7.70 11.60 18.60	4.38 4.39 4.38 4.46	4.38+1.5=5.88 4.39+1.5=5.89 4.38+1.5=5.88 4.46+1.5=5.96	119.12 119.11 119.12 119.04	4.70 4.71 4.70 4.76
	(Sr.no.1+2+3 drop meets the criterion of 0.8% + 0.4% + 3.8% drop as per clause 7.1.9-c, e & f)									
4.	From 125 V DCDB to Loads	3	2C	1 x 2C x 10 1 x 2C x 6 1 x 2C x 4 1 x 2C x 2.5	≤ 355 ≤ 212 ≤ 140 ≤ 88	4.60 7.70 11.60 18.60	4.90 4.90 4.87 4.91	4.90+1.0=5.90 4.90+1.0=5.90 4.87+1.0=5.87 4.91+1.0=5.91	119.10 119.10 119.13 119.09	4.72 4.72 4.70 4.73
	(Sr.no.1+4 drop meets the criterion of 0.8% +4.2% as per clause 7.1.9-c & d)									

Table 29-2: LV Power Copper Conductor Cable Sizes for Various Loads with Different Voltage Drops as per Clause 7.0

Sr. No.	From MDB To	Load		Cable Specification		Actual Length of Cable	Unit Voltage Drop	Actual Voltage Drop	Total Voltage Drop	Voltage Delivered	Voltage Regulation
		VA	A	Type	PVC/PVC XLPE/PVC	(m)	(mV/A/m)	(V)	(V)	(V)	(%)
1.	Station Transformer	500,000	760	1 Core	6x1Cx400 1x1Cx400	≤ 28	0.19/2	2.00	2.00	378.00	0.52
	Amb. 0.95 (Duct) Group 0.87	300,000	456	4 Core	2x4Cx240	≤ 40	0.21/2	1.92	1.92	378.08	0.50
	(Rounded off 2.0 V, the value meets the criterion of 0.8% as per clause 7.1.9.a)										
2.	Battery Charger		150	1 Core	3x1Cx95	≤ 14	0.45	0.95	2.95	377.05	0.77
				4 Core	1x4Cx95	≤ 14	0.44	0.92	2.92	377.08	0.77
	Amb. 1.6 (Air) Group 0.70			1 Core	3x1Cx70	≤ 10	0.60	0.90	2.90	377.10	0.76
				4 Core	1x4Cx70	≤ 10	0.60	0.90	2.90	377.10	0.76
			100	1 Core	3x1Cx70	≤ 15	0.60	0.90	2.90	377.10	0.76
				4 Core	1x4Cx70	≤ 15	0.60	0.90	2.90	377.10	0.76
				1 Core	3x1Cx50	≤ 10	0.84	0.84	2.84	377.16	0.75
				4 Core	1x4Cx50	≤ 10	0.85	0.85	2.85	377.15	0.75
			75	1 Core	3x1Cx50	≤ 15	0.84	0.94	2.94	377.06	0.77
				4 Core	1x4Cx50	≤ 15	0.85	0.95	2.95	377.05	0.77
				1 Core	3x1Cx35	≤ 10	1.13	0.85	2.85	377.15	0.75
				4 Core	1x4Cx35	≤ 10	1.20	0.90	2.90	377.10	0.76
			50	1 Core	3x1Cx35	≤ 16	1.13	0.90	2.90	377.10	0.76
				4 Core	1x4Cx35	≤ 16	1.20	0.96	2.96	377.04	0.78
				1 Core	3x1Cx25	≤ 12	1.55	0.93	2.93	377.07	0.77
				4 Core	1x4Cx25	≤ 12	1.60	0.96	2.96	377.04	0.78
			40	1 Core	3x1Cx25	≤ 15	1.55	0.93	2.93	377.07	0.77
				4 Core	1x4Cx25	≤ 15	1.60	0.96	2.96	377.04	0.78
				1 Core	3x1Cx16	≤ 9	2.50	0.90	2.90	377.10	0.76
				4 Core	1x4Cx16	≤ 9	2.50	0.90	2.90	377.10	0.76
	(By keeping the voltage drop at Sr.no.1 on lower side i.e. 2.0 V instead of 3.0 V, the drop at charger has not been allowed to exceed 0.8 %)										
3.	Air Condensing Unit	63,000	96 (576)*	4 Core	1x4Cx50	≤ 92	0.85	7.50 (45.00)	9.50 (47.00)	370.50 (333.00)	2.50 (12.36)
	Amb. 0.95 (Duct) Group 0.70			4 Core	1x4Cx35	≤ 65	1.20	7.45 (44.90)	9.45 (46.90)	370.55 (333.10)	2.50 (12.35)
		50,000	78 (468)*	4 Core	1x4Cx35	≤ 80	1.20	7.50 (44.90)	9.50 (46.90)	370.50 (333.10)	2.50 (12.35)
				4 Core	1x4Cx25	≤ 60	1.60	7.45 (44.90)	9.45 (46.90)	370.55 (333.10)	2.50 (12.35)
	(*) Starting Values										
	(Sr.no.1+2 drop meets the criterion of 0.8% + 4.2% drop as per clause .1.9.a & b)										

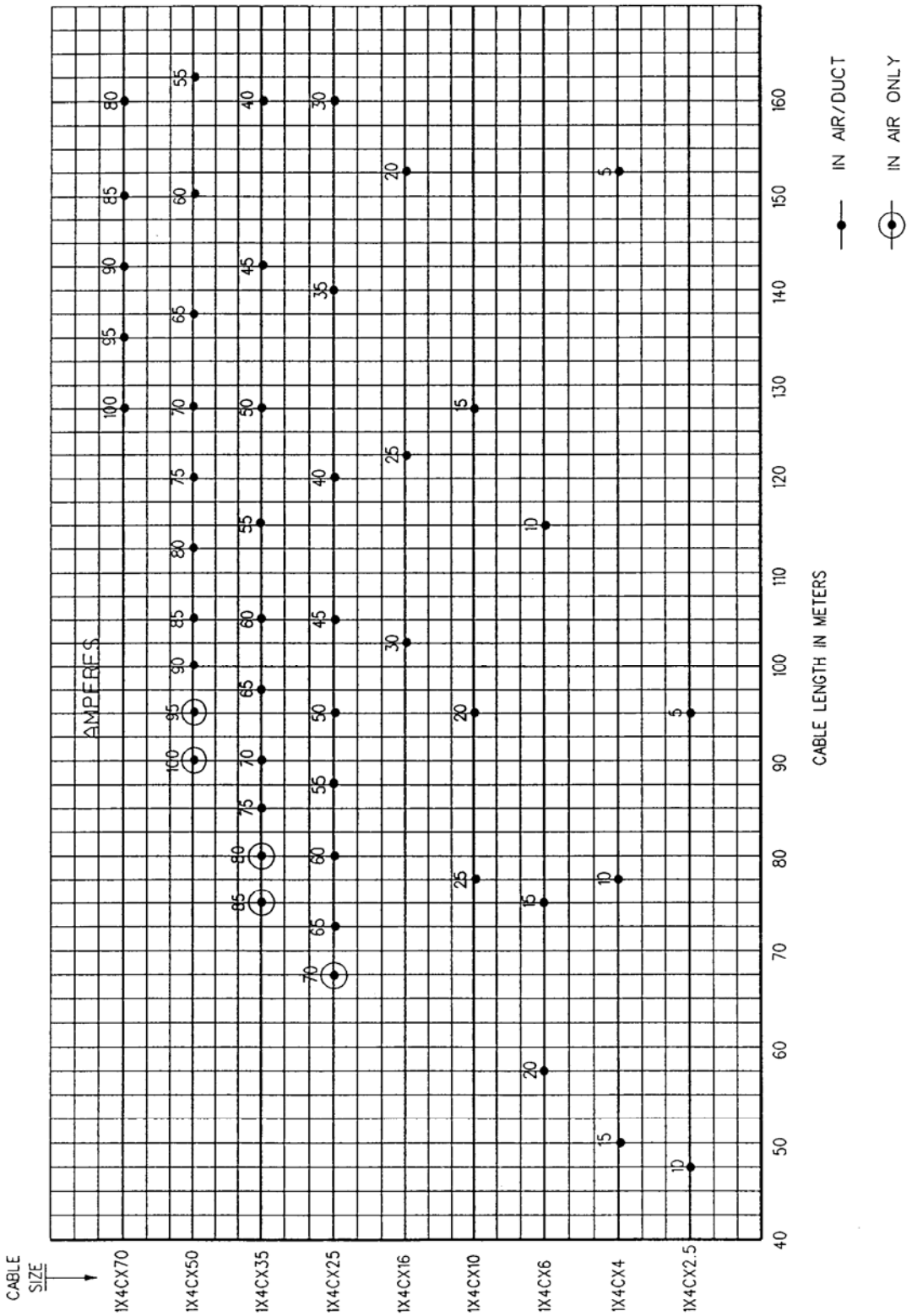

FIGURE 29-1 : CURRENT CARRYING CAPACITY OF AC COPPER CABLES WITH VOLTAGE DROP LIMIT OF 4.2% (16V) (MOTOR LOAD (STARTING 12.5% (47.5V)))

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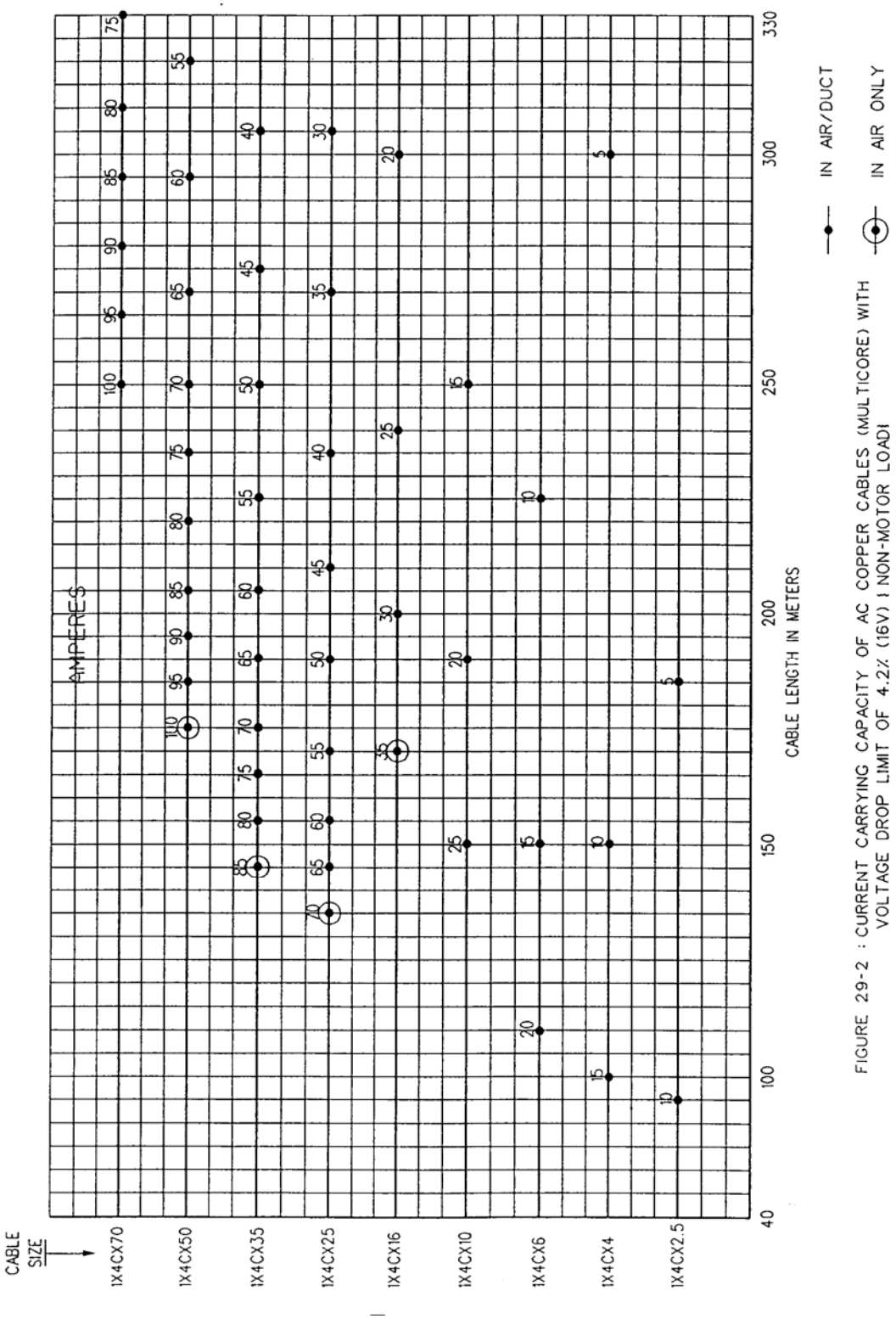

FIGURE 29-2 : CURRENT CARRYING CAPACITY OF AC COPPER CABLES (MULTICORE) WITH VOLTAGE DROP LIMIT OF 4.2% (16V) ! NON-MOTOR LOAD!

DWG # TE-1929-0200-00

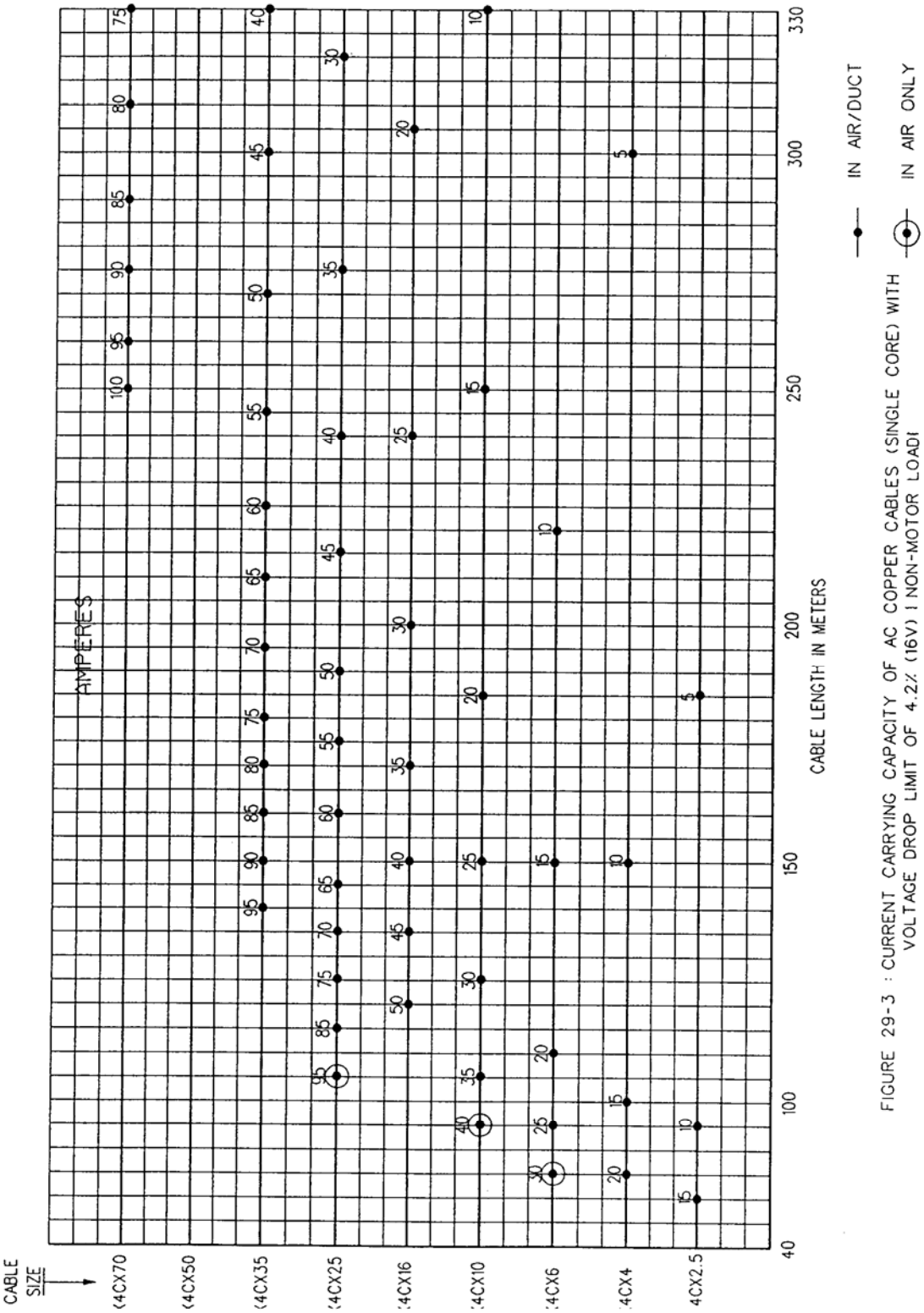

FIGURE 29-3 : CURRENT CARRYING CAPACITY OF AC COPPER CABLES (SINGLE CORE) WITH VOLTAGE DROP LIMIT OF 4.2% (16V) I NON-MOTOR LOAD

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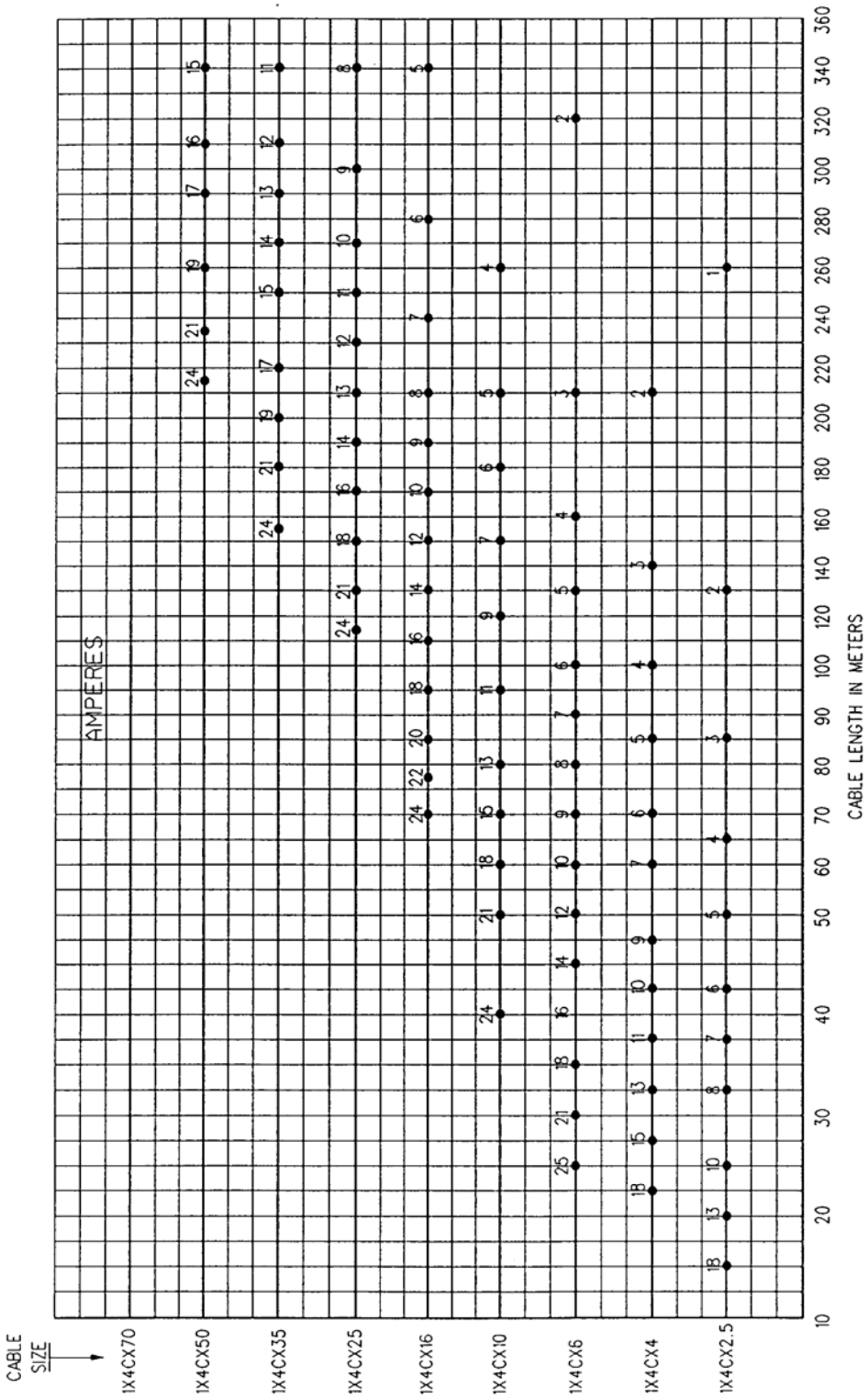

FIGURE 29-4 : CURRENT CARRYING CAPACITY OF DC COPPER CABLES WITH VOLTAGE DROP LIMIT OF 4.2% (5.25V)

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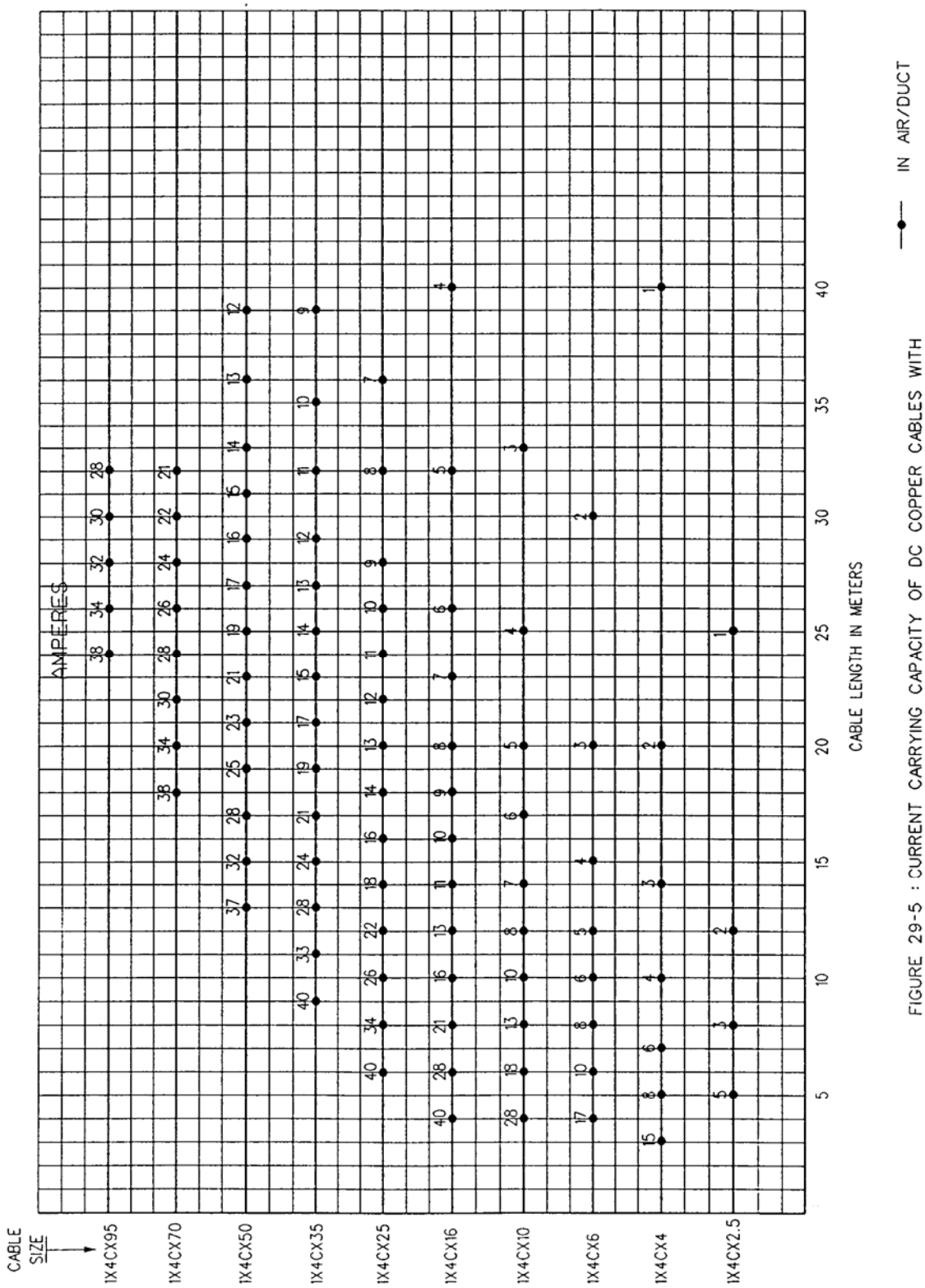

FIGURE 29-5 : CURRENT CARRYING CAPACITY OF DC COPPER CABLES WITH VOLTAGE DROP LIMIT OF 0.4% (0.5V)

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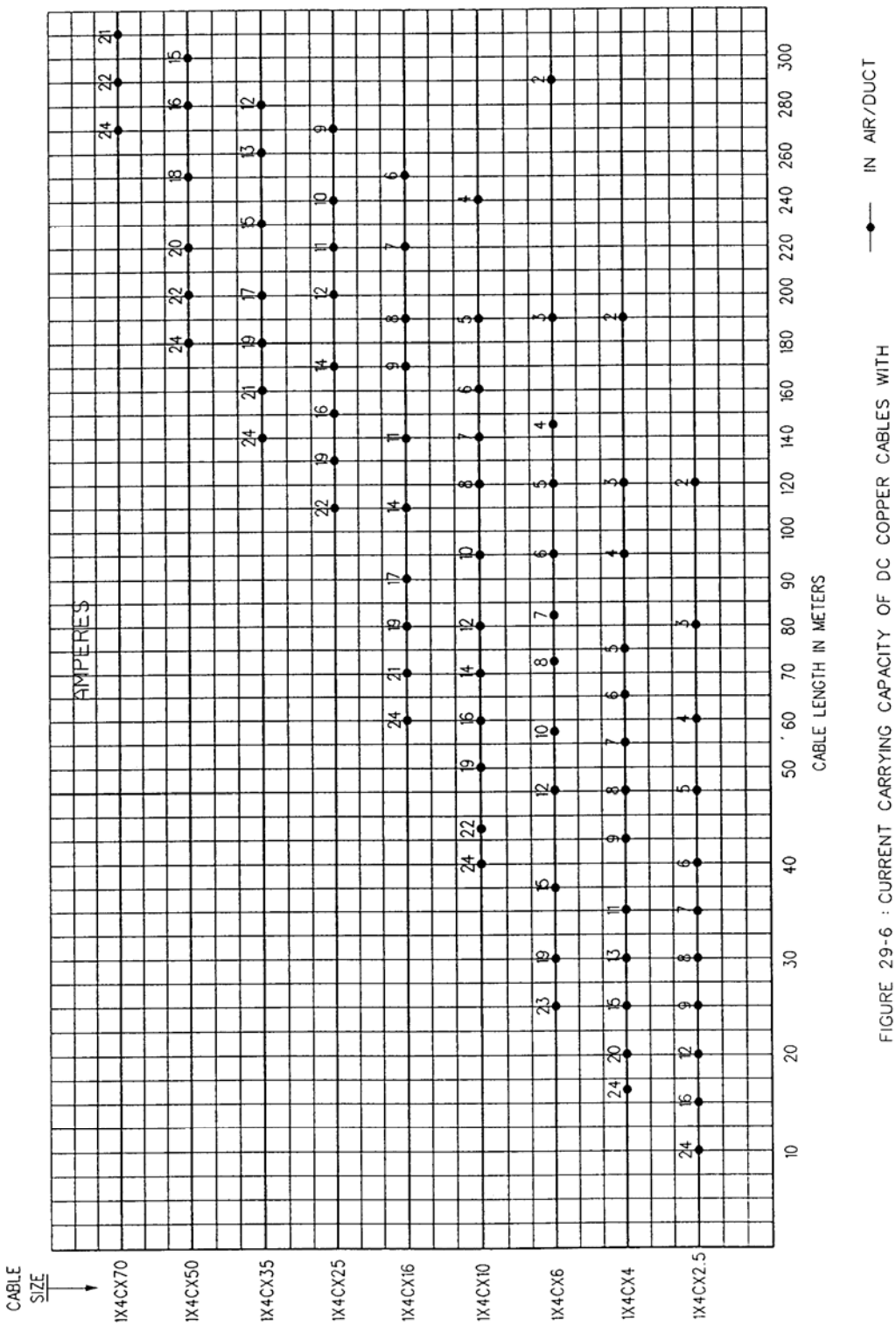

FIGURE 29-6 : CURRENT CARRYING CAPACITY OF DC COPPER CABLES WITH VOLTAGE DROP LIMIT OF 3.8% (4.75V)

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APPENDIX III

AMPACITY AND CORRECTION FACTORS TABLES

This appendix consists of Tables 29-3 to 29-9 and Figure 29-7 providing sample manufacturer's cable ampacity, correction factors resistance and reactances, unit voltage drop and short circuit values.

Table 29-3 : Current Ratings of 600/1000 V - Single and Two Core Copper Conductor PVC Insulated Cable

Nominal Area of Conductor (mm²)	Current Carrying Capacity in Amperes							
	In Air				In Ground		In Duct	
	Single Core		Two Core		Single Core	Two Core	Single Core	Two Core
	Unarmoured	Armoured	Unarmoured	Armoured	Unarmoured & Armoured	Unarmoured & Armoured	Unarmoured & Armoured	Unarmoured & Armoured
1.5	25	-	19	20	30	29	27	23
2.5	34	-	24	28	43	38	36	30
4	46	-	33	37	57	49	48	40
6	57	-	43	47	70	62	62	51
10	76	-	61	65	92	82	83	67
16	97	-	82	86	115	106	107	87
25	132	-	110	115	153	142	139	114
35	164	-	134	141	183	171	163	137
50	199	206	164	171	216	202	191	163
70	252	258	206	215	264	249	232	200
95	310	313	256	265	315	299	273	243
120	361	361	297	308	354	341	302	277
150	413	406	341	352	392	384	332	313
185	477	460	394	409	436	437	367	357
240	568	533	471	487	493	509	410	417
300	656	601	546	563	545	578	451	475
400	765	662	640	658	573	655	483	547
500	889	728	-	-	618	-	518	-
630	1039	804	-	-	663	-	561	-
800	1194	848	-	-	679	-	587	-
1000	1343	907	-	-	709	-	624	-

OPERATING CONDITIONS

1. Continuous operating temperature of conductor 85°C
2. Ambient temperature 52°C
3. Temperature of ground 35°C
4. Depth of Laying in ground 75 cms
5. Thermal resistivity of ground 1.2°C m/w.
6. Single core cables : Two cables, flat formation, laid horizontal with gap = one cable diameter

Table 29-4 : Current Ratings of 600/1000 V - 3 and 4 Core Copper Conductor PVC Insulated Cable

Nominal Area of Conductor (mm²)	Current Carrying Capacity in Amperes			
	In Air		In Ground	In Duct
	Unarmoured	Armoured	Unarmoured & Armoured	Unarmoured & Armoured
	Copper	Copper	Copper	Copper
1.5	15	15	24	20
2.5	21	22	31	27
4	27	29	43	35
6	35	36	53	43
10	49	51	71	58
16	66	68	92	76
25	86	90	120	98
35	106	111	144	117
50	129	134	171	140
70	164	169	210	173
95	202	208	252	207
120	236	241	287	236
150	269	277	320	267
185	310	317	362	302
240	370	374	417	348
300	425	427	467	391
400	492	489	529	447
500	562	554	578	488

OPERATING CONDITIONS

1.	Continuous operating temperature of conductor	85°C
2.	Ambient temperature	52°C
3.	Temperature of ground	35°C
4.	Depth of Laying in ground	75 cms
5.	Thermal resistivity of ground	1.2°C m/w.

Table 29-5: Rating Factors for Variation in Ambient Temperature for Cables Laid in Air

Ambient temperature (°C)	25	30	35	40	45	50	52	55	60
PVC insulated cables	1.35	1.29	1.22	1.16	1.09	1.02	1.00	0.95	0.86
XLPE insulated cables	1.37	1.30	.24	1.17	1.09	1.03	1.00	0.95	0.88

Table 29-6: Rating Factors for Variation in Ground Temperature For Cables Laid Direct in Ground or in Ducts

Ground temperature (°C)	15	20	25	30	35	40	45
PVC insulated cables	1.19	1.15	1.09	1.05	1.00	0.95	0.89
XLPE insulated cables	1.16	1.13	1.08	1.03	1.00	0.95	0.90

Table 29-7: Factors for the Conductor Resistance at Temperatures Other than 20°C

Conductor Temperature	Factor $(1 + \alpha_{20} \cdot \Delta\theta)$		Conductor Temperature	Factor $(1 + \alpha_{20} \cdot \Delta\theta)$	
(°C)	Copper	Aluminum	(°C)	Copper	Aluminum
20	1.0	1.0	65	1.177	1.182
25	1.0196	1.0202	70	1.196	1.204
30	1.0393	1.0403	75	1.216	1.225
35	1.059	1.0604	80	1.236	1.245
40	1.0786	1.0806	85	1.255	1.265
45	1.0982	1.101	90	1.275	1.285
50	1.118	1.121	95	1.293	1.305
55	1.138	1.141	100	1.314	1.325
60	1.157	1.161			

Table 29-8: 600/1000 V Cables AC Resistance and Reactance Values

Nominal Area of Conductor	PVC Insulated Cables					XLPE Insulated Cables				
	A.C. Resistance at 85°C		Reactance at 50 Hz			A.C. Reactance at 90°C		Reactance at 50 Hz		
	Single Core Cables	Multicore Cables	Single Core Cables		Multicore Cables	Single Core Cables	Multicore Cables	Single Core Cables		Multicore Cables
	Copper	Copper	Trefoil	Flat		Copper	Copper	Trefoil	Flat	
(mm²)	(Ohm/km)	(Ohm/km)	(Ohm/km)		(Ohm/km)	(Ohm/km)	(Ohm/km)	(Ohm/km)		(Ohm/km)
1.0	22.72	-	-	-	0.116	-	-	-	-	-
1.5	15.19	15.19	-	-	0.115	15.43	15.43	-	-	0.115
2.5	9.30	9.30	-	-	0.111	9.45	9.45	-	-	0.107
4	5.79	5.79	-	-	0.096	5.88	5.88	-	-	0.093
6	3.87	3.87	-	-	0.092	3.93	3.93	-	-	0.089
10	2.30	2.30	-	-	0.091	2.33	2.33	-	-	0.084
16	1.44	1.44	0.121	0.210	0.086	1.47	1.47	0.114	0.172	0.081
25	0.913	0.913	0.116	0.202	0.085	0.927	0.927	0.113	0.172	0.081
35	0.658	0.658	0.115	0.201	0.083	0.668	0.668	0.110	0.167	0.079
50	0.486	0.486	0.112	0.195	0.080	0.494	0.494	0.106	0.161	0.075
70	0.337	0.337	0.107	0.188	0.077	0.342	0.342	0.103	0.160	0.074
95	0.243	0.243	0.103	0.186	0.077	0.247	0.247	0.098	0.155	0.073
120	0.193	0.193	0.103	0.185	0.075	0.197	0.197	0.097	0.153	0.072
150	0.159	0.159	0.101	0.185	0.075	0.160	0.160	0.097	0.153	0.072
185	0.126	0.126	0.	0.184	0.074	0.128	0.128	0.096	0.153	0.072
240	0.0970	0.0972	0.096	0.181	0.074	0.0986	0.0989	0.092	0.147	0.071
300	0.0785	0.0786	0.094	0.180	0.073	0.0800	0.0802	0.090	0.147	0.070
400	0.0628	0.0630	0.091	0.178	0.073	0.0640	0.0645	0.090	0.146	0.070
500	0.0515	0.0518	0.089	0.176	0.072	0.0525	0.0530	0.089	0.146	0.070
630	0.0426	-	0.086	0.173	-	0.0428	-	0.086	0.144	-
800	0.0357	-	0.086	0.171	-	0.0380	-	0.084	0.143	-
900	0.0310	-	0.084	0.168	-	0.0334	-	0.081	0.140	-

Table 29-9: 600/1000 V PVC Insulated Cables Voltage Drops at Maximum Conductor Operating Temperature in V/A/km

Nominal Area of Conductor (mm²)	3 or 4 Single Core Cables		2 Core or 2 Single Core Cables - Touching	3 and 4 Core Cables
	Copper Trefoil	Copper Flat Spaced *	Copper	Copper
1.5	26.3	26.3	30.4	26.3
2.5	16.1	16.1	18.6	16.1
4	10.0	10.0	11.6	10.0
6	6.7	6.7	7.7	6.7
10	4.0	4.0	4.6	4.0
16	2.5	2.5	2.9	2.5
25	1.55	1.60	1.8	1.6
35	1.13	1.16	1.3	1.2
50	0.84	0.87	0.98	0.85
70	0.60	0.65	0.69	0.60
95	0.45	0.52	0.51	0.44
120	0.37	0.45	0.41	0.36
150	0.31	0.41	0.35	0.30
185	0.26	0.38	0.29	0.25
240	0.23	0.35	0.24	0.21
300	0.20	0.33	0.21	0.18
400	0.19	0.32	0.19	0.16
500	0.18	0.31	0.17	0.15
630	0.16	0.31	0.16	-
800	0.16	0.30	0.15	-
1000	0.15	0.29	0.15	-

* Adjacent cables spaced by one cable diameter

Maximum conductor operating temperature for PVC insulated cables 85°C

$$I_{SC} = 0.11 \times \frac{A}{\sqrt{t}}$$

where

I_{SC}	Short Circuit Current in kA
A	Conductor Area in mm^2
t	Short Circuit Time in seconds

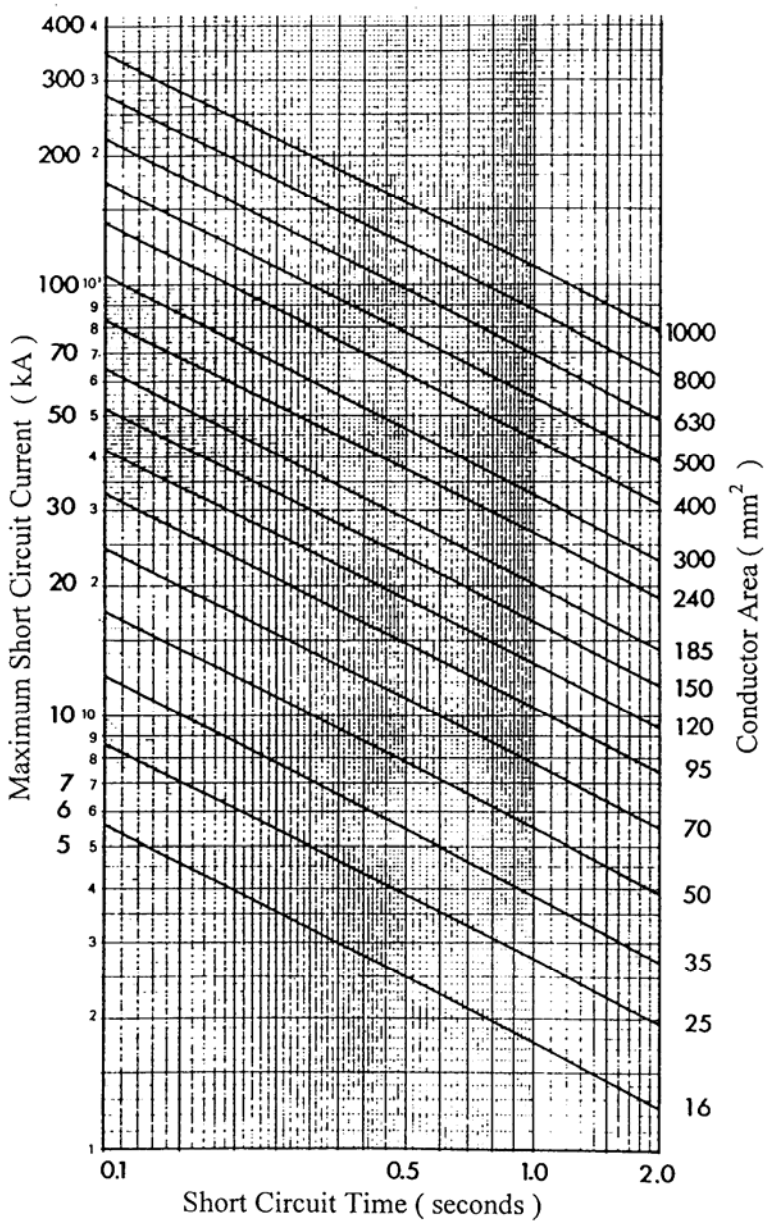

Figure 29-7 : Short Circuit Current Curves for Copper Conductor, PVC Insulated Cables

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